McGill-FIAM ASSET MANAGEMENT HACKATHON 2026

Market-neutral U.S. equities

Four research pods, one fixed blend, beta-neutral by construction

Applications of Machine Learning in Portfolio Management

STOCHASTIC PIRATES FUND

Walk-forward test 01/2021-08/2026 | Net of costs and borrow | October 2026

Four research pods, 15.6% a year compared a 7.5% hurdle

15.6%
CAGR, net; hurdle 7.5%, S&P 500 13.5%

0.80
Information ratio vs T-bill + 4%

1.22
Sharpe ratio, excess of T-bill

12.4%
CAPM alpha per year (t = 3.1, HAC)

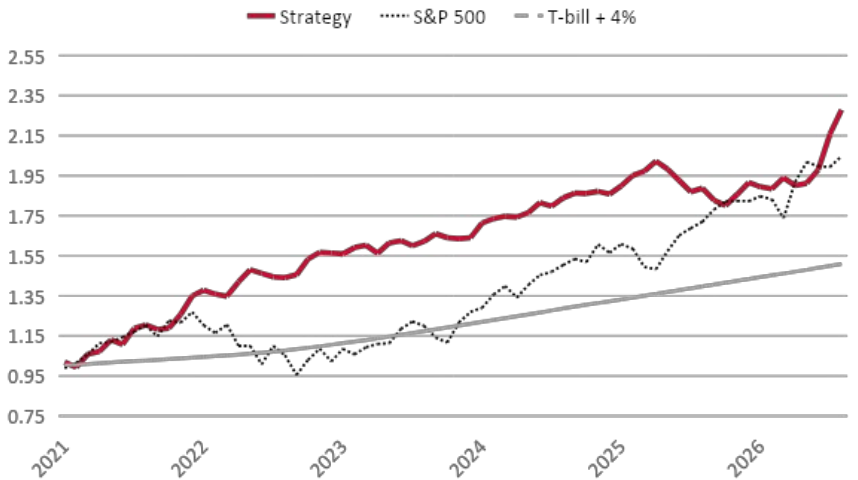
-0.06
Monthly beta to S&P 500 (SE 0.08); daily -0.02

-11.0%
Max drawdown, monthly; S&P 500 -24.8%

Summary Table (Arcana)

% Idio (absolute)	14.2%	Gross | AUM | %	$3,169 | $2,183 | 145.15%
		Net ($M) | %	$979 | 45%
% Idio+Industry of Book Var	33.8%	Long Exposure ($M) | %	$2,074 | 95%
Predicted Beta | $M	-0.05x | -$118	Short Exposure ($M) | %	-$1,095 | -50%
Predicted Beta (Longs) | $M	0.59x | $1,228	Portfolio Vol ($M) | Daily	$177.6 | $11.2
Predicted Beta (Shorts) | $M	-1.23x | -$1,346	Portfolio Vol (%) | Daily	8.1% | 0.5%
Effective | Total Positions	149.5 | 200	Value at Risk (VaR) | Daily	13.4% | 0.8%
Effective | Total Long Positions	90.0 | 100	Top 10 % of Total Variance	28%
Effective | Total Short Positions	59.5 | 100	Top 10 % of Idio Variance	23%
PF Sharpe EV at 53 % Hit, 10 x Turns	1.17x	Portfolio IRR	NA

Growth of $1, 01/2021-08/2026 (net)



Four pods. Complementary signals. One fixed blend

What each pod reads

Pod	Economic idea / signal	Share	CAGR
Fundamentals	How is the business performing? Slow expectation updates: earnings, revisions, sales vs. inventory	49%	7.6%
Valuation	What is it worth? Value vs. price: fair-value gap and cash runway	14%	2.2%
Events	What is changing? Disclosure complexity: categories and frequency of 8-K events	20%	3.1%
Dynamics	How do stocks behave? Relative-return patterns across 147 company characteristics	17%	2.7%
All four	Fixed rank blend + light peer-graph tilt	100%	**15.6%**

Correlation of monthly pod returns

	Fundamentals	Valuation	Events	Dynamics
Fundamentals	1.00	-0.47	-0.46	-0.54
Valuation	-0.47	1.00	+0.49	+0.56
Events	-0.46	+0.49	1.00	+0.75
Dynamics	-0.54	+0.56	+0.75	1.00

Fundamentals offsets the other pods (-0.46 to -0.54).

Blend: 15.6% CAGR vs. 5.8-11.0% standalone.

From economic hypothesis to tested signal

Papers & data	Economic hypothesis	Measurable signal	Evidence gate	Decision
LLM-assisted search of dated papers and alternative data	State the economic mechanism before testing	Use information available before the investment period	t-stat >= 2; shuffled control Correlation limit; score floor	Admit or reject

321 -> 12

Candidate variants -> admitted
Recovered agent research portfolio

Holding periodle	Scheduled LLM	Stated cutoff
2021-2022	GPT-4	Sep 2021
2023	GPT-4o	Oct 2023
2024	o3	Jun 2024
Jan-Jun 2025	GPT-5	Sep 2024
Jul-Dec 2025	GPT-5.4	Aug 2025
Jan-Apr 2026	GPT-5.5	Dec 2025
May-Aug 2026	GPT-6 Astra	Apr 2026

Signal / source	Entered	Exited	Avg gross / mo.
Cash value quality · LLM	Apr 2021	Feb 2023	+0.54%
Liquid reversal · LLM	Apr 2021	Dec 2021	+2.45%
Earnings surprise / weak response: Paper, Ye & Schuller 2020	Apr 2022	Jan 2023	+0.46%
Profitable capex: LLM	Apr 2023	Nov 2023	+1.00%
Foot traffic vs. sales: Alt data, owned extract	Apr 2024	Sep 2024	+2.35%
Gross profit / conservative assets: LLM	Jul 2024	Dec 2024	-0.44%
Sales value / equity-sale filings: Alt data, SEC 8-K	Oct 2024	Active Jun 2026	+2.42%
Gross profitability: Paper, Novy-Marx 2013	Jan 2022	Jul 2022	-0.12%
Asset growth: Paper, Cooper et al. 2008	Oct 2022	Jun 2023	-0.06%
Price momentum: Paper, momentum research	Jan 2023	Jun 2023	+0.90%
Sales growth vs. SG&A: Paper, Abarbanell & Bushee 1998	Oct 2023	Jan 2026	+0.97%
Short-sale volume: Alt data, FINRA	Jul 2024	Aug 2025	+0.30%

Alternative data tested or explored: foot traffic; card spending / user breadth; website visits; app downloads; SEC 8-K, Form 4 and 13F; FINRA short-sale volume; Polymarket; Kalshi; job openings; trade claims.

Data and method: monthly ranks, annual walk-forward

Data

- **Core:** 147 monthly characteristics and 8-K filings, Jan 2015-Aug 2026. Daily returns from 2014.
- **I/B/E/S:** revisions for one disclosed alpha sleeve.
- **Carbon Arc:** no predictive asset in this book.
- **Reporting:** FRED T-bill + 4% hurdle and S&P 500. Arcana beta informs sizing.

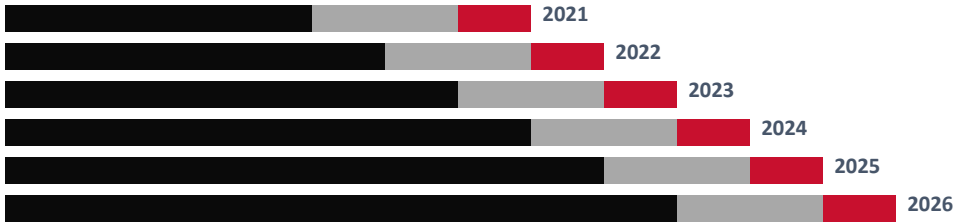
Data screening: findings and gaps

- **History:** supplied inputs start before 2021. I/B/E/S bounds remain unverified.
- **Lag:** month t predicts t+1. Filings enter from the next trading day.
- **Revisions:** historical vintages remain unverified.
- **Coverage:** 8-K survivorship bias. Universe coverage and return-bias tests remain open.

Models and training

- **Signals:** rules and boosted trees rank next-month excess returns in four pods.
- **Blend:** average available sleeves, plus a 5% graph rank tilt where available.
- **Annual refit:** train through Y-3. Validate on Y-2/Y-1. Test on Y.
- **Research agents:** propose sleeves annually. Only prior-year gate passers enter the blend.

Walk-forward scheme, annual refit



Black: expanding training window from 2015

Grey: two pre-test validation years

Red: test years, Jan 2021-Aug 2026

2021: train 2015-18 / validate 2019-20

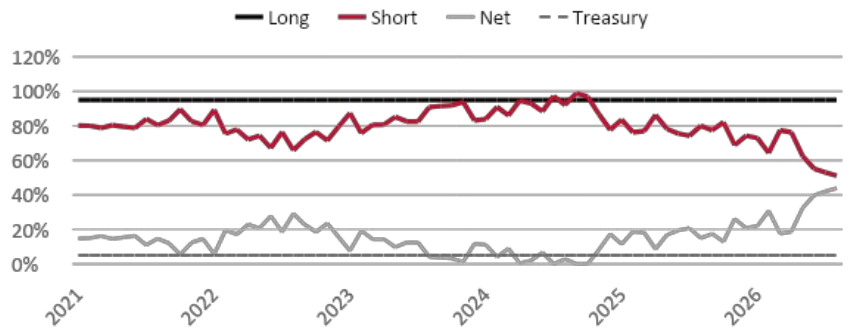
2026: train 2015-23 / validate 2024-25

Saved replay reuses frozen forecasts.

Construction: 95% long, variable short, beta constrained at formation

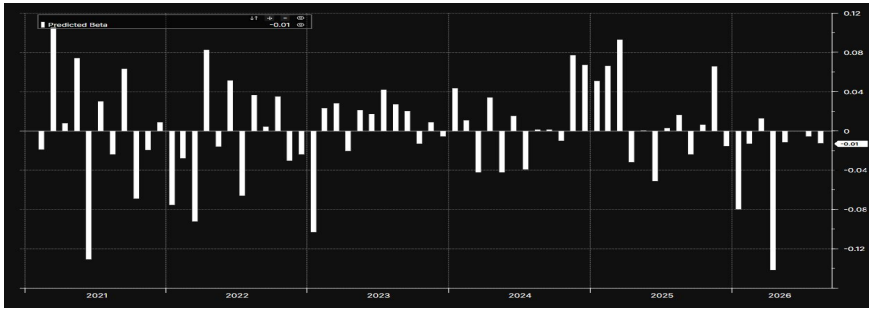
Rank	Select	Screen	Size	Rebalance
Average the pod ranks into one score	100 long, 100 short, sector-balanced; incumbents kept within a 0.5 rank buffer	$5 price; shorts ≥ $500m cap; trades ≤ 5% (long) / 2% (short) of 5-day volume	Beta within ±0.025 on 3 estimators; gross ≤ 200%, net ±50%, sectors ±20%	Monthly; one-way turnover < 400% per calendar year

Capital at each rebalance, % of NAV



Cash is explicit: 5% in T-bills every month; short proceeds earn about 2.6% of NAV a year as collateral. Shorts carry higher beta, so ~80% short offsets 95% long; net dollars moved from 5% (2024) to 31% (2026).

Rolling 252-day beta to the S&P 500



At formation, local daily and monthly beta and the Arcana/local hybrid beta each remain within ±0.025. Realised monthly beta is -0.06; daily beta is indicative.

Risk is controlled by fixed rules, not by agents

Screen	Size	Protect	Constrain	Check
$5 price; shorts ≥ $500m cap; trades ≤ 5% (long) / 2% (short) of 5-day volume	100 long, 100 short; each position ≤ 5% of NAV; 95% long, 5% T-bills	Riskiest 5% of shorts by peer squeeze odds halved; rallied shorts cut to ≤ 1.5× target	|β| ≤ 0.025 on 3 estimators; gross ≤ 200%, net ±50%, sectors ±20%, ≥ 5 sectors	Every month tested and logged; a sector-rule breach stops the build

Each rule, its limit, and what it did over 68 months

Control	Level	Limit	Realised
Liquidity screen	Position	$5 price; short cap ≥ $500m	~1,600 short names excluded a month; smallest short $513m
Position size	Position	≤ 5% of NAV	Largest position 5.0%
Squeeze (peer EVT)	Position	Riskiest 5% of shorts halved	13 shorts in 7 months
Run-up cap	Position	Short ≤ 1.5× its target	122 shorts trimmed, in 42 of 68 months
Beta	Book	|β| ≤ 0.025, 3 estimators	Max 0.025 at formation; realised monthly -0.06
Exposure	Book	Gross ≤ 200%, net ±50%, sector ±20%	Max 194% / 44% / 12.6%
Turnover	Book	< 400% per calendar year	391% annualised

Agents sit upstream (slide 4)

They only propose new signal sleeves before each test year. 144 proposed, 0 passed the gate, so no agent sets a weight or a limit.

Agent vs fixed rule

0 admissions: both runs hold the same book. Worked example: the best of 51 proposals on 2015-2020 reached spread t = 1.05 vs t ≥ 2, so it was rejected.

Expected-shortfall and edge throttles are logged monthly but do not bind under the fixed 95% long deployment. Daily volatility is 9.8% on available days vs the 12% cap; formally unverified until daily held returns are complete.

Performance: full out-of-sample period

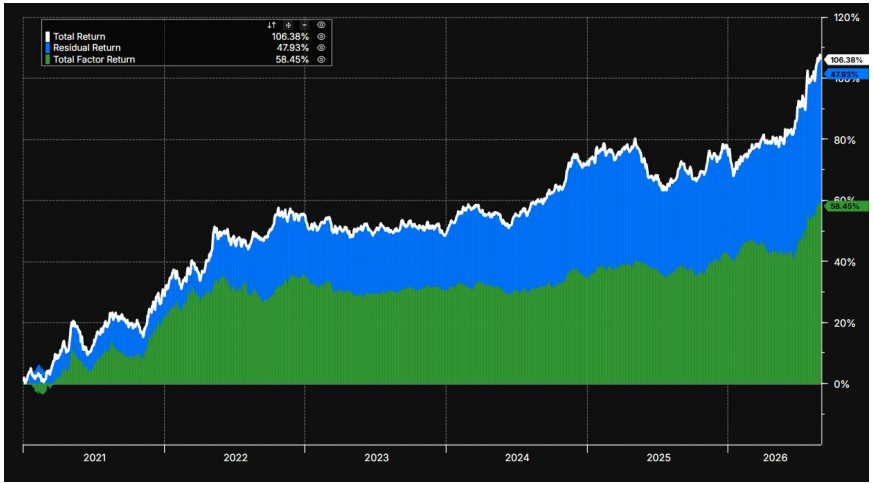
Metric	Strategy	Benchmark
Ending endowment ($bn)	2.279	1.509
Cumulative return (including T-bills)	127.89%	50.87%
Arithmetic annual return	15.06%	7.28%
CAGR	**15.65%**	**7.53%**
Volatility, monthly ann.	**9.54%**	**0.56%**
Alpha vs S&P 500 (s.e.)	12.36% (4.16%)	–
Monthly beta (s.e.)	-0.055 (0.079)	–
Daily beta (s.e.), indicative	-0.023 (0.016)	–
Sharpe	1.22	–
IR vs T-bill + 4%	0.80	–
Tracking error	9.67%	–
Max drawdown, monthly	-10.96%	0.00%
Worst month	-3.21%	0.34%

Metric	Strategy
Worst day, indicative	-2.99% (2021-06-02)
Positions avg / min / max	200 / 200 / 200
Gross avg / max	174.93% / 193.93%
Net avg / min / max	15.07% / -3.93% / 43.90%
One-way turnover / rebalance	32.62%
One-way turnover, annualised	391.43%
Holding period, estimated	126 trading days
Average position	$14.10m
Short median market cap	$15,533m
Annual trading cost	0.76% of NAV
Annual borrow cost	0.44% of NAV
4-factor alpha (t, HAC)	5.84% (0.94)
Long / short leg P&L, p.a.	+16.2% / -3.1%

Local no-agent hybrid replay, monthly, net of modeled costs. Daily figures are indicative pending NAV reconciliation.

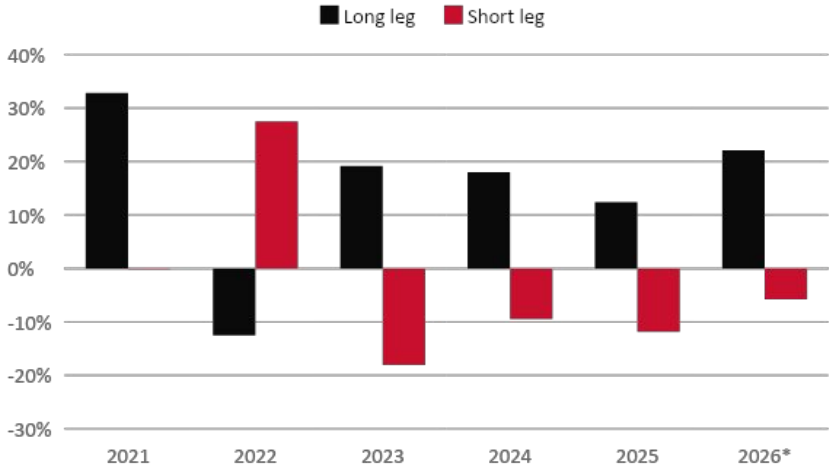
Where the edge is: stock selection, mostly on the long side

Idiosyncratic vs factor return (Arcana)



Arcana attributes 47.93 percentage points to residual and 58.45 to factors over the period; gross of local modeled costs.

P&L by leg and year, % of NAV

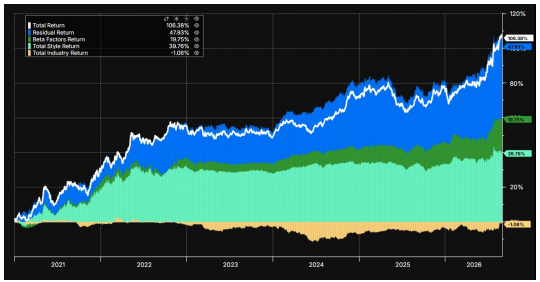


Longs earn, shorts insure: the long leg made +16.2% of NAV a year, positive in 5 of 6 years. The short leg cost -3.1% a year but paid +27.5% in 2022, when longs lost 12.6%.

Arcana Portfolio 73, historical, before modeled costs. Leg P&L before costs, borrow and cash income; *2026 = January-August.

Arcana: return sources and risk

Decomposition



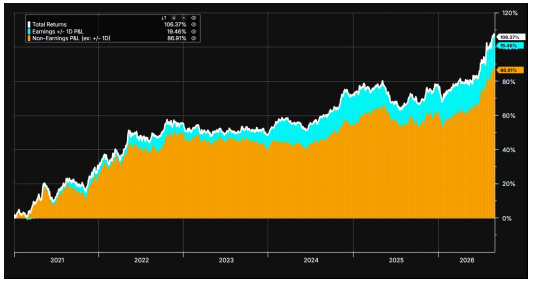
Long vs short spread



Concentration



Earnings P&L



Arcana: exposures and trading

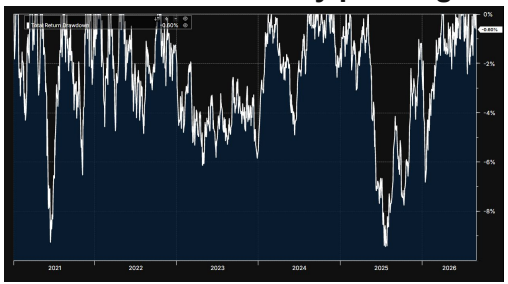
Summary: VaR and volatility

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% Idio+Industry of Book Var	33.8%	Long Exposure ($M) | %	$2,074 | 95%
Predicted Beta | $M	-0.05x | -$118	Short Exposure ($M) | %	-$1,095 | -50%
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Effective | Total Positions	149.5 | 200	Value at Risk (VaR) | Daily	13.4% | 0.8%
Effective | Total Long Positions	90.0 | 100	Top 10 % of Total Variance	28%
Effective | Total Short Positions	59.5 | 100	Top 10 % of Idio Variance	23%
PF Sharpe EV at 53 % Hit, 10 x Turns	1.17x	Portfolio IRR	NA

Style exposure



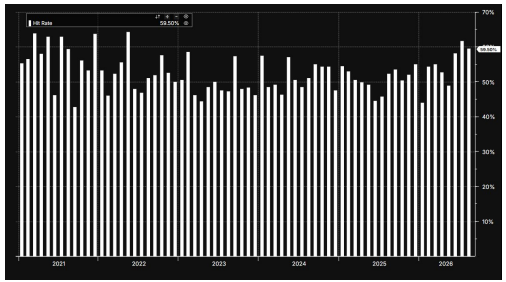
Drawdown: S&P overlay pending



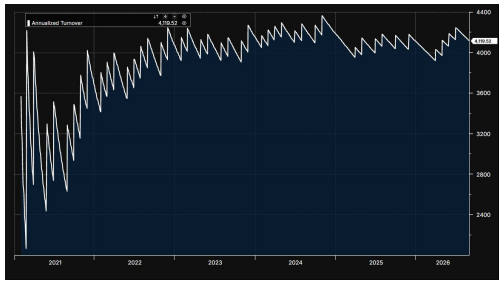
Sharpe ratio



Hit rate



Turnover



Discussion: what worked and what comes next

What worked

– Performed as trained: rank IC 0.108 a month, positive in 5 of 6 years

– 2022 rate shock: shorts in speculative growth (QS, QuantumScape; PLUG, Plug Power) paid +27.5% of NAV

– Longs in the 2025-26 capex cycle: FIX, Comfort Systems USA (+3.1% of NAV); NEM, Newmont (+2.9%)

What did not

– Crowded thematic shorts squeezed: OKLO, Oklo; GEV, GE Vernova; AMC, AMC Entertainment Holdings

– 2023 and 2025 mega-cap rallies: short leg -18.0% and -11.9% of NAV

– Part of the alpha is size and value exposure

– Agents: 144 proposals, 0 admitted

Next steps

– Cap crowded thematic shorts; test a squeeze filter

– Reconcile daily and monthly NAV; certify daily volatility

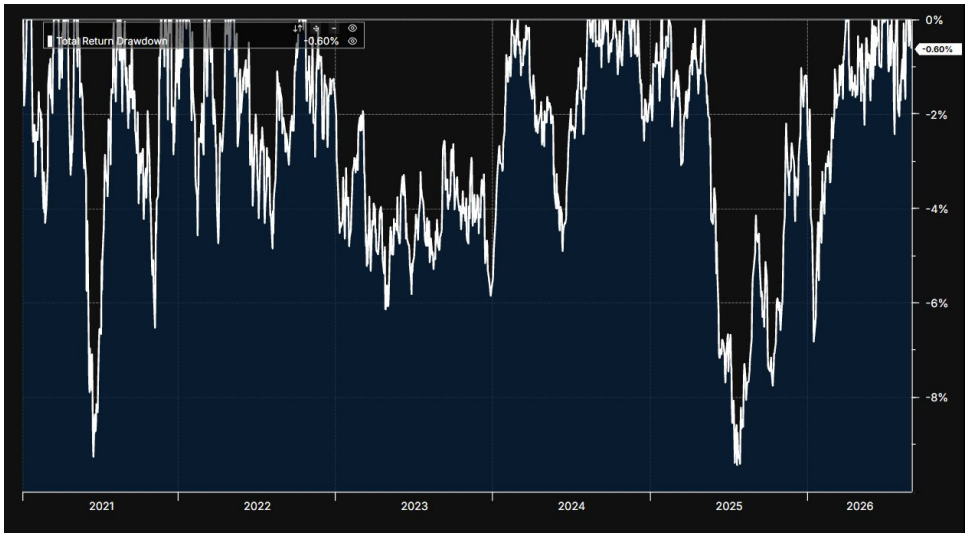
– Compare agent vs fixed rule across regimes

– Cost sensitivity at 2x modeled costs

Costs: 17.0% gross CAGR and 15.6% net CAGR. Modeled annual trading cost 0.76% of NAV; borrow 0.44%.

Appendix A1. where we lose, how we protect, how we come back

Daily drawdown from peak (Arcana)



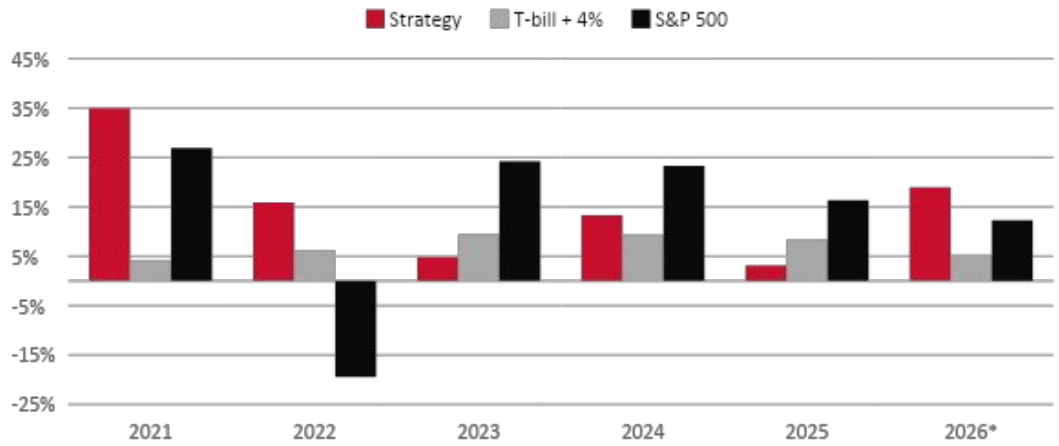
History: +35.0% in 2021, +15.9% in 2022 while the S&P fell 19.4%, +4.8% in 2023, +13.3% in 2024, +3.1% in 2025, +18.9% in 2026 to August.

Risk discussion: the short book lost during the 2023 and 2025 rallies. The local replay shows a monthly maximum drawdown of 11.0%. The supplied Arcana chart does not include the S&P 500 benchmark.

Arcana drawdown screenshot; S&P 500 benchmark overlay still needed. Local daily NAV reconciliation remains pending.

Appendix A2. Calendar-year returns

Calendar-year returns, net



	2021	2022	2023	2024	2025	2026*
Strategy	35.0%	15.9%	4.8%	13.3%	3.1%	18.9%
T-bill + 4%	4.1%	6.2%	9.4%	9.3%	8.4%	5.2%
S&P 500	26.9%	-19.4%	24.2%	23.3%	16.4%	12.3%
Active vs hurdle	+30.8%	+9.7%	-4.6%	+4.0%	-5.2%	+13.7%

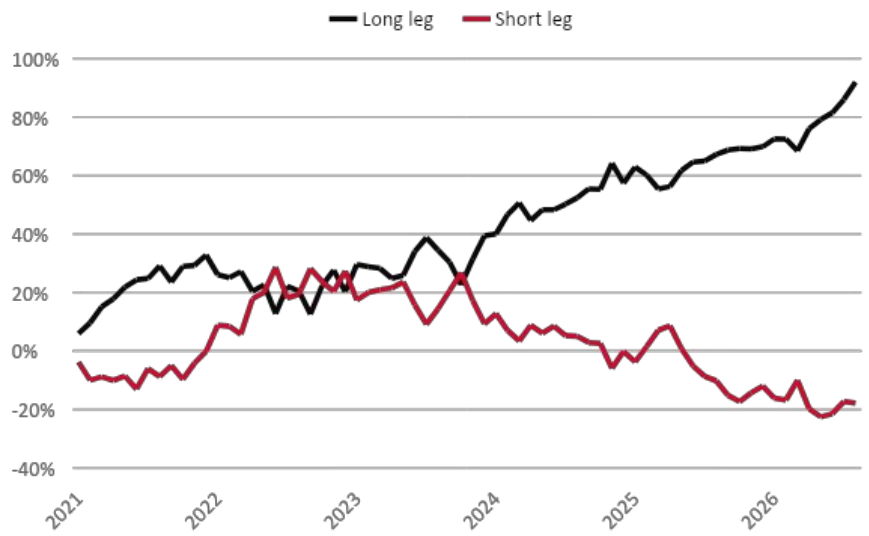
Monthly NAV, net of trading costs (modeled) and borrow fees; *2026 = January-August. Hurdle = 3-month T-bill + 4%.

01/2021-08/2026	Strategy	Hurdle	S&P 500
CAGR	15.6%	7.5%	13.5%
Cumulative	+127.9%	+50.9%	+104.6%
Volatility, ann.	9.5%	0.6%	15.1%
Sharpe	1.22	n/a	0.70
Information ratio	0.80	n/a	0.44
Max drawdown	-11.0%	0.0%	-24.8%
Best month	+8.9%	+0.8%	+10.4%
Worst month	-3.2%	+0.3%	-9.3%
Hit rate vs hurdle	54%	n/a	59%

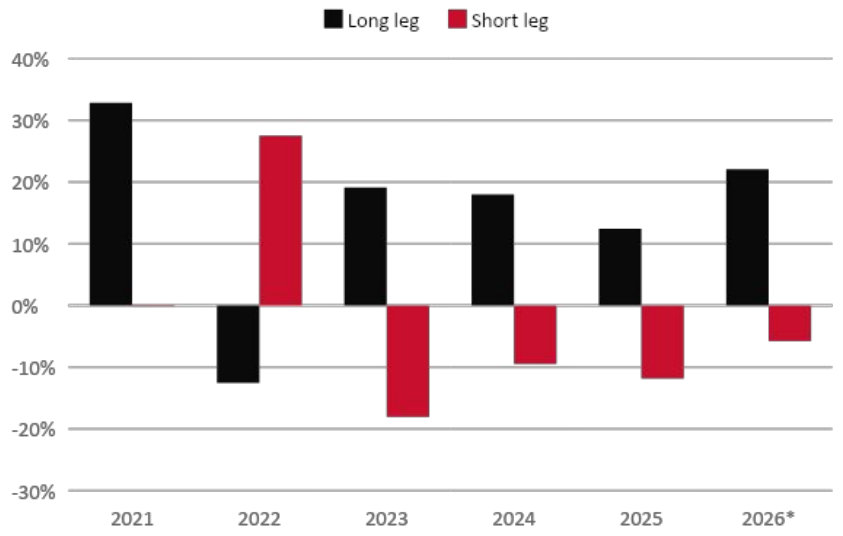
Read-across. Strongest when the market fell or broadened (2021, 2022, 2026); weakest in the narrow mega-cap rallies of 2023 and 2025, when the short book lost.

Appendix A3. Long and short leg P&L

Cumulative P&L by leg, % of NAV (simple sum)



P&L by leg and year, % of NAV



The long leg is the alpha engine: +16.2% of NAV a year, positive in five of six years. **The short leg is the hedge:** -3.1% a year on average, but +27.5% in 2022 when longs lost 12.6%. Its cost shows up in strong rallies (2023: -18.0%, 2025: -11.9%).

Price P&L before trading costs, borrow fees and cash income. *2026 = January-August.

Appendix A4. Our max drawdown came in a rising market

Drawdown from peak, month-end (net)



05/2025-10/2025	May	Jun	Jul	Aug	Sep	Oct	Total
Fund, net	-2.0%	-2.9%	-2.9%	+1.0%	-3.2%	-1.5%	-11.0%
Long leg	+5.5%	+2.9%	+0.4%	+2.2%	+1.5%	+0.5%	+13.0%
Short leg	-7.8%	-5.9%	-3.6%	-1.5%	-5.0%	-2.2%	-25.9%
S&P 500	+6.2%	+5.0%	+2.2%	+1.9%	+3.5%	+2.3%	+22.8%

Monthly net NAV; leg P&L is price P&L before costs and cash income. Stock returns are cumulative excess returns while shorted.

Why we fell while the market rose

A neutral book loses when shorts beat longs, not when the market falls. From the April 2025 peak to October, the S&P 500 rebounded 22.8% from the tariff sell-off; our longs made +13.0% of NAV, but our shorts lost 25.9%.

A speculative rally. The steepest gains came in the names we were short: OKLO, Oklo (+450%); APLD, Applied Digital (+241%); QUBT, Quantum Computing (+143%). Speculative, high-beta themes outran our longs.

Not a beta miss. Formation beta stayed between -0.02 and +0.04 every month: the loss was style, not market direction.

2022 in reverse. The same short leg made +27.5% of NAV when the market fell 19.4%.

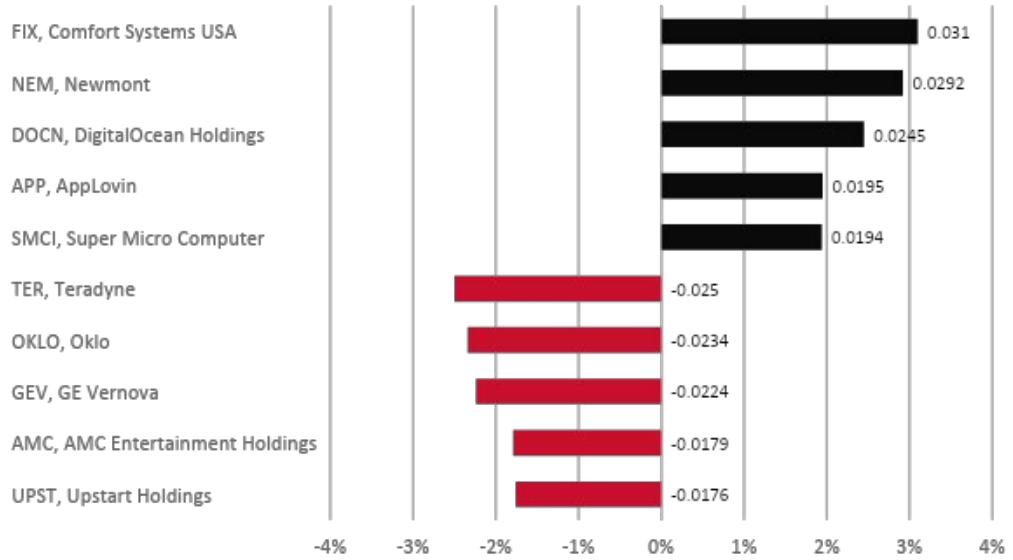
Response: run-up and squeeze limits on shorts were in force but did not prevent it; next is a cap on crowded thematic shorts. Recovered to a new high by 07/2026.

Appendix A6. Implementation and capacity

Metric	Value
One-way turnover, full years	395% (cap < 400%)
Trading cost, p.a.	0.76% of NAV
Borrow cost, p.a.	0.44% of NAV
Average / max position	0.87% / 5.0%
Top-10 names, share of gross	10.3%
Short book, median market cap	$15.5bn
Short book, avg dollar volume	$506m / day
Median position, days of volume	0.13 long / 0.07 short
Est. holding period	126 trading days

Capacity. Every trade is under 5% (long) or 2% (short) of five days of volume; the median position is a fraction of one day's volume.

Five largest gains and losses, cumulative % of NAV



No single name decides the result: the ten best added 21% of NAV and the ten worst cost 18%, against +127.9% cumulative. Losses cluster in crowded themes (nuclear, AI infrastructure, meme stocks).

Contributors: cumulative price P&L while held, % of NAV, 01/2021-08/2026. Full top/bottom ten in Appendix A7.

Appendix A7. Largest holdings and contributors

Top 10 longs, avg weight % NAV

Ticker	Company	Wt	Mo.
CAT	Caterpillar	+1.04	33
DBX	Dropbox	+0.89	53
MCK	Mckesson	+0.75	35
CROX	Crocs	+0.70	33
FIX	Comfort Systems Usa	+0.67	19
MELI	Mercadolibre	+0.67	35
MPC	Marathon Petroleum	+0.66	40
DUOL	Duolingo	+0.66	40
ABNB	Airbnb	+0.66	32
TSN	Tyson Foods	+0.61	38

Top 10 shorts, avg weight % NAV

Ticker	Company	Wt	Mo.
APO	Apollo Global Manageme	-0.97	66
MRVL	Marvell Technology	-0.67	44
BA	Boeing	-0.61	42
ENPH	Enphase Energy	-0.58	43
RH	R H	-0.55	38
MSTR	Strategy	-0.53	41
DLTR	Dollar Tree	-0.53	35
SAIA	Saia	-0.51	42
EQT	EQT	-0.49	47
CTLT	Catalent	-0.48	39

10 largest gains, % NAV

Ticker	Company	P&L
FIX	Comfort Systems Usa	+3.10
NEM	NEWMONT	+2.92
DOCN	DigitalOcean	+2.45
APP	Applovin	+1.95
SMCI	Super Micromputer	+1.94
QS	Quantumscape	+1.72
PLUG	Plug Power	+1.71
RNG	RingCentral	+1.66
MU	Micron Technology	+1.60
NUE	Nucor	+1.56

10 largest losses, % NAV

Ticker	Company	P&L
TER	Teradyne	-2.50
OKLO	Oklo	-2.34
GEV	GE Vernova	-2.24
AMC	A M C Entertainment	-1.79
UPST	Upstart	-1.76
AAOI	Applied Optoelectronic	-1.58
APLD	Applied Digital	-1.51
FSLR	First Solar	-1.50
SNAP	Snap	-1.42
APO	Apollo Global Manageme	-1.37

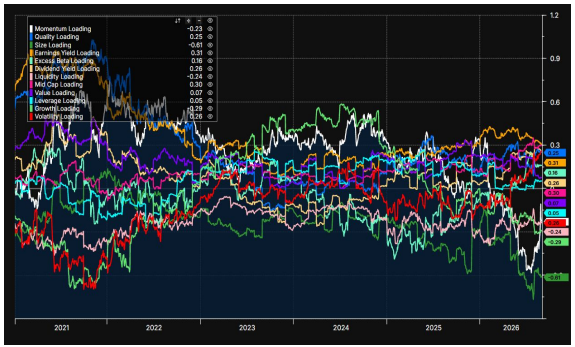
Company labels from 8-K registered symbols, SEC issuer names and CRSP history (holdings_label_sources.csv).

Appendix A8. Further Arcana analytics

Long vs short book



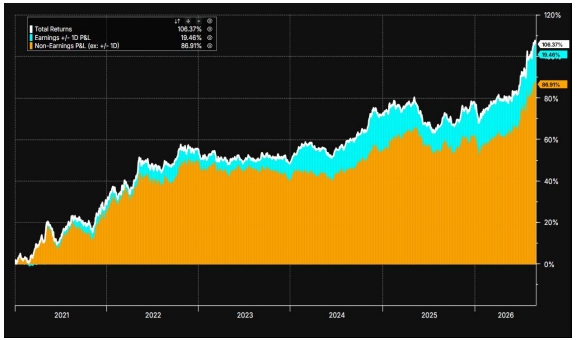
Style factor loadings



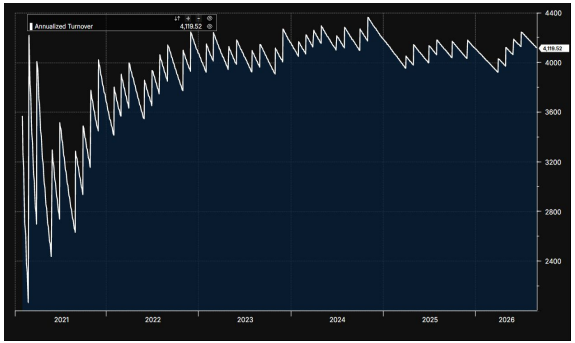
Concentration



Earnings-event P&L



Annualised turnover



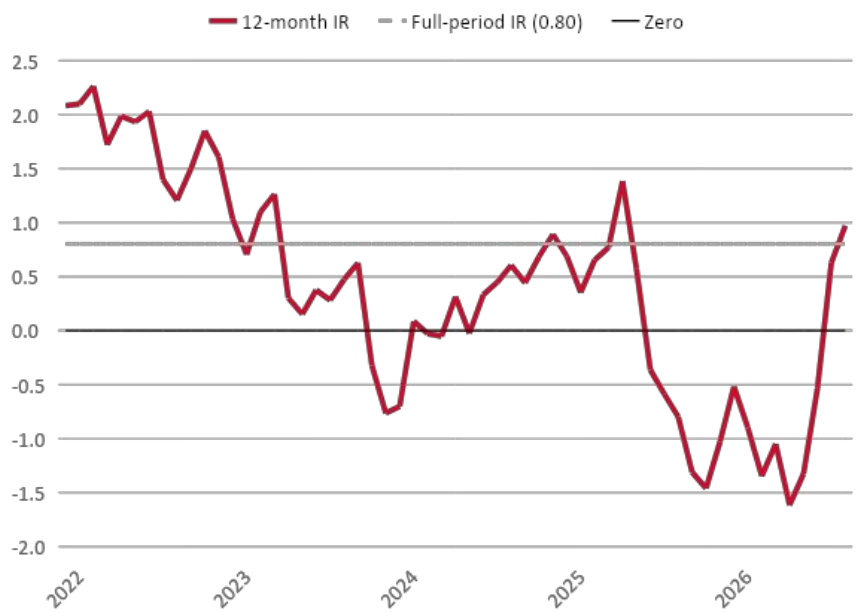
Realised vs predicted volatility



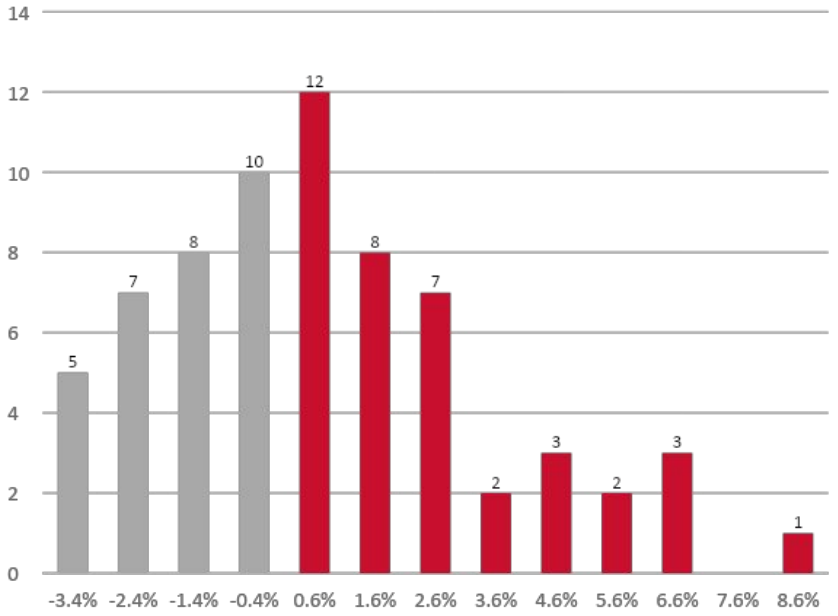
Arcana Portfolio 73; historical output, 01/2021–08/2026. Volatility image still requires updating.

Appendix A9. Rolling information ratio and return distribution

Rolling 12-month information ratio vs T-bill + 4% (net)



Monthly returns, count of months (net), 1-pt bins



Red bars beat the hurdle (0.61% a month, T-bill + 4%); grey bars fall short. Bin labels are the lower edge.

Monthly net returns, 01/2021-08/2026; rolling window needs 12 months, so it starts 12/2021. The 12-month IR swings with the 2023 and 2025 rallies; the full-period IR is 0.80.

Appendix A10. Data screening and limitations

History

Supplied monthly characteristics start in 2015. Confirm the first usable date for each I/B/E/S and 8-K input.

Publication lag

Inputs are joined at month t for returns at t+1. Verify each source's actual release and availability lag.

Revisions and vintages

Whether each revised input preserves historical vintages remains unverified. Daily and monthly NAV do not reconcile.

Coverage and bias

The 8-K corpus has survivorship limits and I/B/E/S coverage ends early 2026. Universe coverage and return-related bias require measurement.

Appendix A11: Beta estimation and monitoring

How we got there

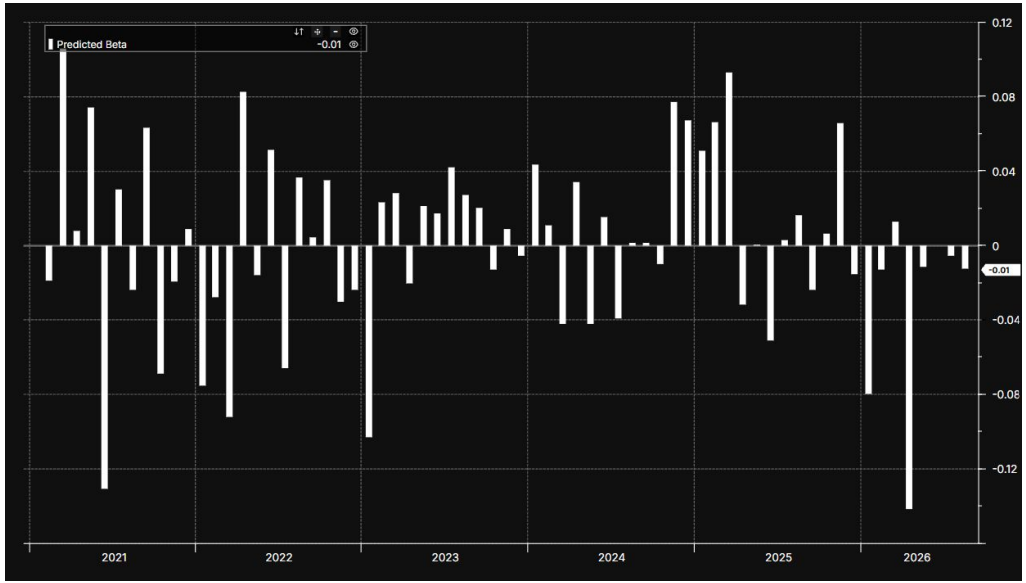
1. Three estimators. Local daily beta (252 sessions), local monthly beta (36 months) and the Arcana US model beta to SPY, with local fallback where Arcana has no prior-month value.

2. One hard limit. The sizing solver keeps the portfolio beta within ±0.025 on all three at every formation, then a guard re-checks it.

3. Monitor after. We track realised beta on daily and monthly returns; drift between rebalances is the residual risk.

Realised. Monthly beta -0.06 (SE 0.08); daily beta -0.02 (SE 0.016); correlation with the large-cap market -0.09.

Rolling 252-day beta to the S&P 500 (month-end)



Daily series starts once 252 sessions are available (01/2022). Daily NAV reconciliation is pending; the monthly estimate is the certified one.